

Hölder Fixed-Point Retractions from Strong Convexity

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Status and source question

This is a `candidate_partial_likely_valid` result for the final problem in Andrzej Wiśnicki, *Hölder continuous retractions and amenable semigroups of uniformly Lipschitzian mappings in Hilbert spaces*, arXiv:1204.6464 [1]. The source recalls the Goebel–Kirk theorem: if X is uniformly convex, $C \subset X$ is bounded, closed, and convex, $T : C \rightarrow C$ is uniformly k -Lipschitzian, and

$$k(1 - \delta_X(1/k)) < 1,$$

then T has a fixed point. On PDF page 6 it asks whether the fixed-point set is a Lipschitz or Hölder retract. The displayed problem writes $\text{Fix } \mathcal{S}$ although the referenced theorem concerns one map; the surrounding discussion makes $\text{Fix } T$ the natural single-map reading.

In general, the conditions “ $RUC(S)$ has a left invariant mean” and “ S is left reversible” are independent of each other. However, if $RUC(S)$ has sufficiently many functions to separate closed sets, then the former condition implies the latter (see, e.g., [12, Lemma 3.1]).

Recently, Górnicki (cf. [9, Cor. 14]) proved that if S is left reversible and $\mathcal{S} = \{T_t : t \in S\}$ is a uniformly k -Lipschitzian semigroup on C , then the set of fixed points of $\mathcal{S}$ is a retract of C . It is not clear how to extend Theorem 2 to left reversible semigroups. There are many other open problems in this area. Perhaps the most natural is the following.

Problem. Under the assumptions of Theorem 1, is the fixed-point set $\text{Fix } \mathcal{S}$ a Lipschitz or Hölder continuous retract of C ?

The arbitrary uniformly convex case is not settled here. We prove a quantitative strong-convexity criterion and obtain a genuine non-Hilbert subcase in L^p , $3/2 < p < 2$.

Strong convexity and proof intuition

Assume that X is reflexive and that for some $c \in (0, 1]$ and every $v \in X$ there is $j(v) \in X^*$ such that

$$\|u\|^2 \geq \|v\|^2 + 2\langle j(v), u - v \rangle + c\|u - v\|^2 \quad (u \in X). \quad (1)$$

This says that the squared norm has Bregman strong-convexity constant c . In a Hilbert space $c = 1$ and (1) is the variance identity with a nonnegative remainder. In L^p , $1 < p \leq 2$, one may take $c = p - 1$ by the sharp Ball–Carlen–Lieb inequality [2].

For a left amenable semigroup, average an orbit using an invariant mean and call its vector barycenter Lx . Unlike an asymptotic-center map, L is automatically k -Lipschitz. Inequality (1) replaces the exact Hilbert variance identity and proves that the variance about the next barycenter contracts by

$$\rho = \frac{k^2 - c}{c}.$$

When $k^2 < 2c$, the iterates of L have geometrically decaying increments. Their uniform limit is Hölder because L^n is k^n -Lipschitz.

Main partial theorem

Theorem 1 (strongly convex amenable actions). *Let X be a reflexive Banach space satisfying (1), let C be a nonempty bounded closed convex subset of X , and let S be a discrete left amenable semigroup. Suppose $\mathcal{S} = \{T_s : s \in S\}$ acts on C , with $T_{st} = T_s T_t$, and is uniformly k -Lipschitzian:*

$$\|T_s x - T_s y\| \leq k \|x - y\| \quad (s \in S, x, y \in C).$$

If $k^2 < 2c$, then the common fixed-point set

$$\text{Fix } \mathcal{S} = \{x \in C : T_s x = x \text{ for every } s \in S\}$$

is nonempty and is a Hölder continuous retract of C .

Proof. Let μ be a left invariant mean on $\ell^\infty(S)$. Reflexivity and the Riesz-style barycenter construction used in [1] define $Lx \in C$ by

$$\langle Lx, x^* \rangle = \mu_t \langle T_t x, x^* \rangle \quad (x^* \in X^*). \quad (2)$$

Indeed Lx lies in the weakly closed convex hull of the orbit. Moreover, duality immediately gives

$$\|Lx - Ly\| \leq k \|x - y\|. \quad (3)$$

Set

$$\mathcal{Q}(x) = \mu_t \|T_t x - Lx\|^2, \quad \mathcal{E}(x) = \mu_t \|T_t x - x\|^2.$$

Fix $s \in S$. Apply (1) with

$$u = T_{st}x - T_s Lx, \quad v = Lx - T_s Lx,$$

and then take the mean in t . Left invariance makes the barycenter of $(T_{st}x)_t$ equal to Lx , so the linear term has mean zero and $\mu_t \|T_{st}x - Lx\|^2 = \mathcal{Q}(x)$. The Lipschitz bound for T_s therefore gives

$$\|T_s Lx - Lx\|^2 + c\mathcal{Q}(x) \leq \mu_t \|T_{st}x - T_s Lx\|^2 \quad (4)$$

$$\leq k^2 \mu_t \|T_t x - Lx\|^2 = k^2 \mathcal{Q}(x). \quad (5)$$

Taking the mean in s yields

$$\mathcal{E}(Lx) \leq (k^2 - c)\mathcal{Q}(x). \quad (6)$$

Apply (1) once more to the orbit $(T_s Lx)_s$, whose barycenter is $L^2 x$, now with reference point Lx . Again the linear term vanishes:

$$\mathcal{E}(Lx) \geq \|L^2 x - Lx\|^2 + c\mathcal{Q}(Lx). \quad (7)$$

Combining (6) and (7) gives

$$\mathcal{Q}(Lx) \leq \rho \mathcal{Q}(x), \quad \rho = \frac{k^2 - c}{c} < 1. \quad (8)$$

The case $\rho = 0$ makes Lx fixed after one step and is immediate, so assume $0 < \rho < 1$.

Write $x_m = L^m x$ and $D = \text{diam } C$. Jensen's inequality for the mean, (6), and (8) show, for $m \geq 1$, that

$$\|x_{m+1} - x_m\|^2 \leq \mathcal{E}(x_m) = \mathcal{E}(Lx_{m-1}) \quad (9)$$

$$\leq (k^2 - c)\mathcal{Q}(x_{m-1}) \leq (k^2 - c)\rho^{m-1}D^2. \quad (10)$$

Thus $L^m x$ converges uniformly in x to a map $Rx \in C$.

Put $\gamma = \sqrt{\rho}$. From (10), the tail after the n -th iterate is $O(\gamma^n)$, while (3) gives $\|L^n x - L^n y\| \leq k^n \|x - y\|$. Balancing these two bounds proves that R is Hölder. Explicitly, for $k > 1$ one may take

$$\alpha = \frac{\log(1/\gamma)}{\log k + \log(1/\gamma)} \in (0, 1), \quad (11)$$

and when $k = 1$, the uniform limit is nonexpansive.

It remains to identify the range. From (6) and (8), $\mathcal{E}(x_m) \rightarrow 0$. For every t ,

$$\|T_t Rx - Rx\|^2 \leq 3k^2 \|Rx - x_m\|^2 + 3\|T_t x_m - x_m\|^2 + 3\|x_m - Rx\|^2.$$

Taking the mean and then letting $m \rightarrow \infty$ gives $\mathcal{E}(Rx) = 0$. For fixed $s \in S$, the triangle inequality and left invariance give

$$\begin{aligned} \|T_s Rx - Rx\|^2 &\leq 2\mu_t \|T_s Rx - T_t Rx\|^2 + 2\mathcal{E}(Rx) \\ &= 2\mu_t \|T_s Rx - T_{st} Rx\|^2 + 2\mathcal{E}(Rx) \\ &\leq 2(k^2 + 1)\mathcal{E}(Rx) = 0. \end{aligned}$$

Hence $Rx \in \text{Fix } \mathcal{S}$. If $x \in \text{Fix } \mathcal{S}$, then (2) gives $Lx = x$, and consequently $Rx = x$. Thus R is the asserted retraction. $\square$

Corollary 2 (L^p subcase). *Let $3/2 < p \leq 2$, let C be a nonempty bounded closed convex subset of a real L^p space, and let a discrete left amenable semigroup act on C by uniformly k -Lipschitzian maps. If*

$$k < \sqrt{2(p-1)},$$

then its common fixed-point set is a Hölder retract of C . In particular, the same conclusion holds for the fixed-point set of one uniformly k -Lipschitzian map, by applying the result to its power semigroup.

Proof. For $1 < p \leq 2$, the Ball–Carlen–Lieb inequality says

$$\frac{\|a + b\|_p^2 + \|a - b\|_p^2}{2} \geq \|a\|_p^2 + (p-1)\|b\|_p^2.$$

Equivalently, $x \mapsto \|x\|_p^2$ is strongly convex. Passing from the midpoint inequality to the standard two-point inequality and differentiating at one endpoint gives (1) with $c = p-1$. Theorem 1 applies. The strict range is nonempty for $k \geq 1$ precisely when $p > 3/2$. $\square$

Scope, verification, and remaining obstruction

Corollary 2 gives a non-Hilbert family of affirmative cases. For the source’s exact formulation, one may additionally impose its Goebel–Kirk inequality; Theorem 1 itself does not need that separate fixed point criterion.

The full problem for arbitrary uniformly convex spaces remains open in this packet. The published asymptotic-center construction gives geometric convergence, but its continuity is governed by δ_X^{-1} near zero. A uniformly convex norm need not have a power-type modulus, so that argument does not automatically give a Hölder exponent. The invariant-mean method above instead stops at $k^2 = 2c$. No claim is made that this threshold is sharp.

The included regression script checks the Ball–Carlen–Lieb midpoint inequality on randomized finite-dimensional ℓ^p vectors and audits the sign of ρ throughout sampled admissible ranges. It is not used as proof.

Eight focused attempts are recorded in the accompanying attempt note. Bounded run-index and web/arXiv searches on 17 August 2026 used the exact source problem, the paper title and author, and the phrases *uniformly Lipschitzian fixed point Hölder retract*, *L^p Hölder retract*, and *invariant mean strong convexity uniformly Lipschitzian*. They found continuous-retract results, the Hilbert theorem, and asymptotically regular variants, but no explicit statement of this criterion. Novelty confidence is moderate pending specialist review.

Human-review recommendation. Check the passage from the Ball–Carlen–Lieb midpoint inequality to (1), and the use of a finitely additive invariant mean in the two strong-convexity averages. Also search the nonlinear ergodic literature for an equivalent variance-contraction criterion.

References

- [1] A. Wiśnicki, *Hölder continuous retractions and amenable semigroups of uniformly Lipschitzian mappings in Hilbert spaces*, Topol. Methods Nonlinear Anal. 43 (2014), 89–96; arXiv:1204.6464.
- [2] K. Ball, E. A. Carlen, and E. H. Lieb, *Sharp uniform convexity and smoothness inequalities for trace norms*, Invent. Math. 115 (1994), 463–482.
- [3] E. Sędlak and A. Wiśnicki, *On the structure of fixed-point sets of uniformly Lipschitzian mappings*, Topol. Methods Nonlinear Anal. 30 (2007), 345–350.